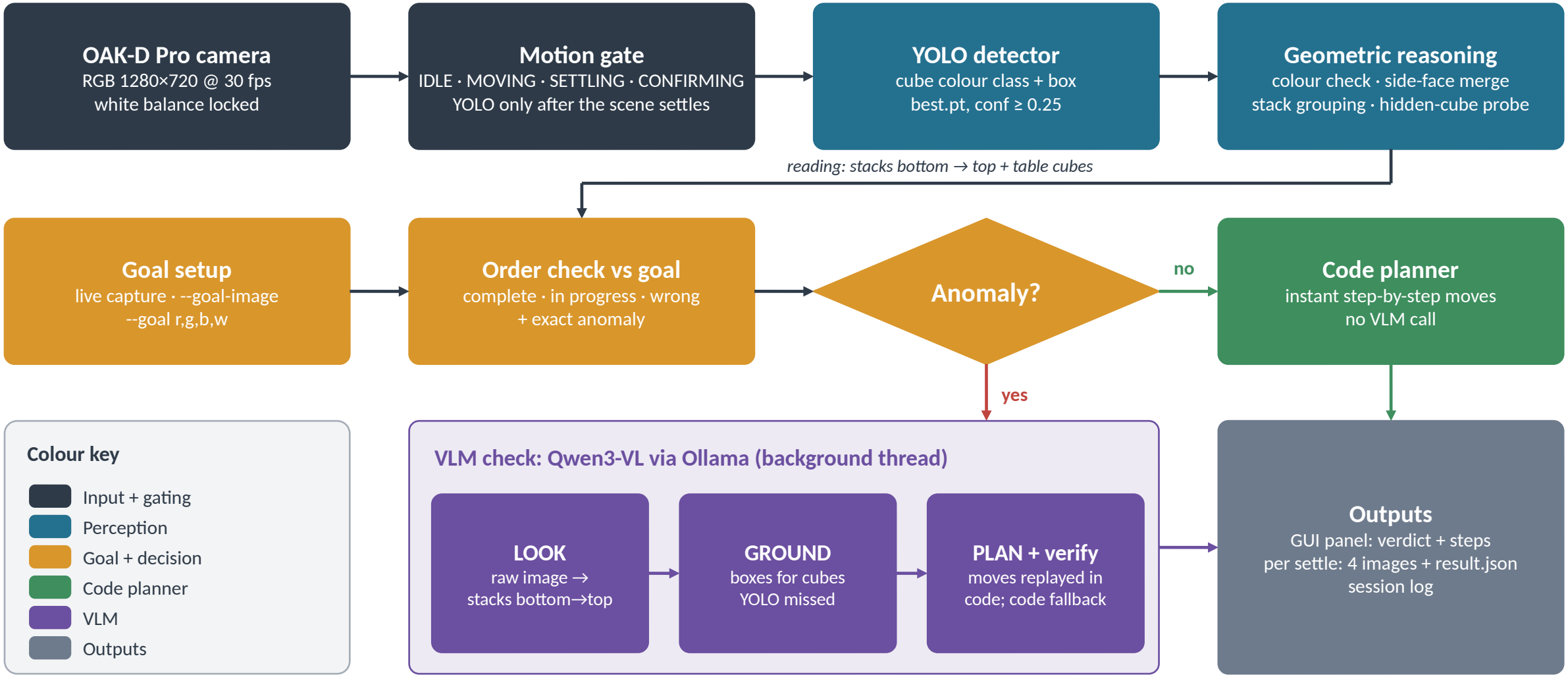
Cube stacking assistant

System architecture: motion-gated YOLO + geometric reasoning, a goal set from an image, and a VLM (Qwen3-VL via Ollama) called only on anomalies



goal, bottom to top

End-to-end architecture



Every settle is saved (images + JSON) whether or not the VLM is called. The camera loop never waits for the VLM.

Start-up and goal setup



1 Default: live capture

- Window shows "SET THE GOAL"
- Build the target stack, press SPACE
- Reading shown with boxes: ENTER accepts, SPACE captures again

2 --goal-image FILE

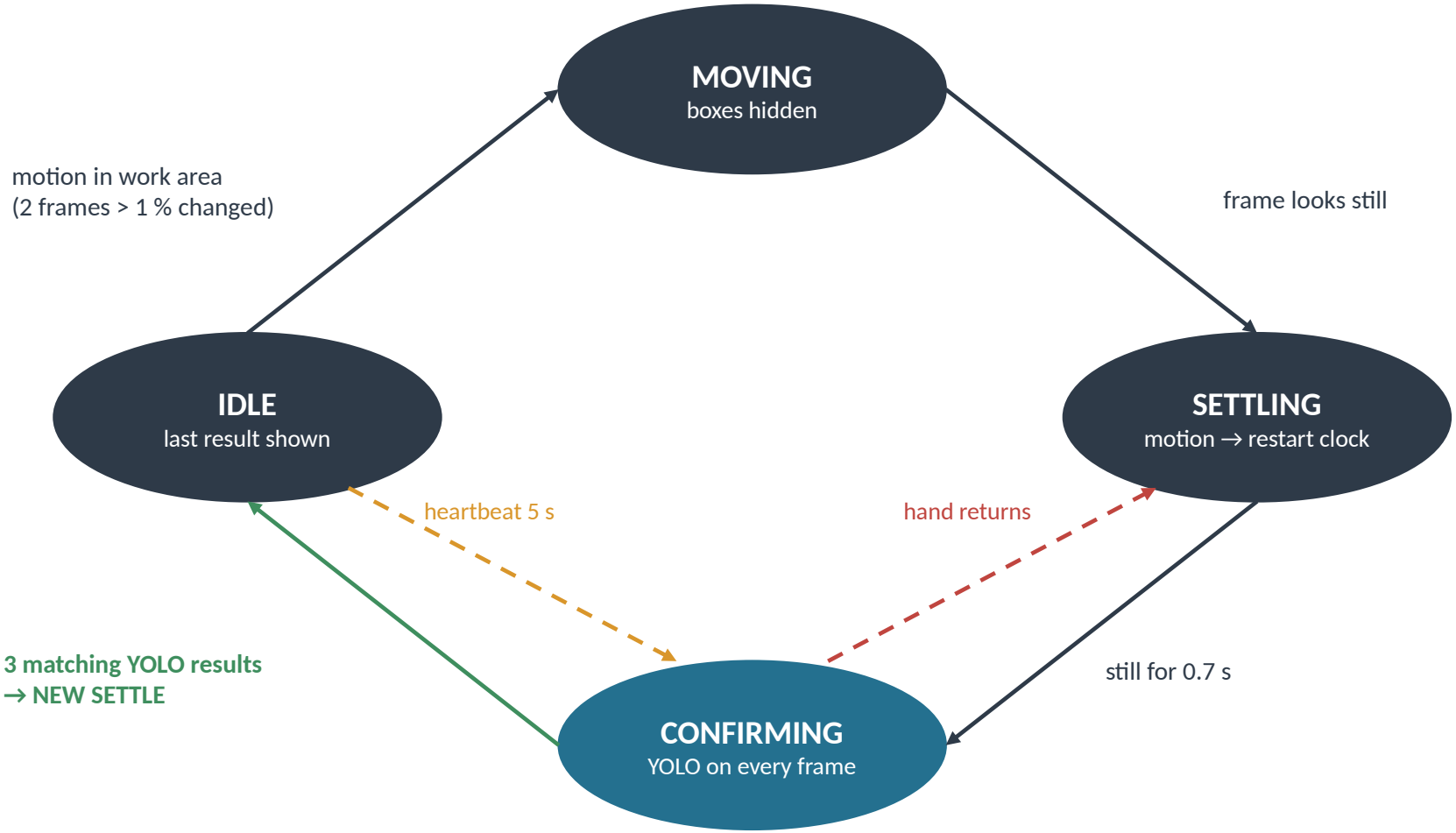
- Photo of the target stack
- Read once at start, no confirmation
- Needed with --image (single-image mode)

3 --goal red,green,black,white

- Order given directly, bottom → top
- Colours checked against the YOLO classes
- No photo needed

How a goal photo is read: YOLO + geometric reasoning, then the VLM LOOK on the raw image; the tallest stack is the goal. Saved in result/goal/goal_<time>/ (photo, annotated photo, goal.json).

Motion gate: run YOLO only when the scene settles



How it works

- Motion = colour change against an anchor frame, in the work area set with the a key
- Comparing with an anchor, not the last frame, catches a slow hand
- CONFIRMING ends when 3 YOLO readings agree (or the majority after 15)
- Every settle after real motion is saved, even if the reading did not change
- A heartbeat counts only if the reading changed

Perception: YOLO + geometric reasoning

1 YOLO detector

Class (red / green / black / white) and box for each visible cube. Often misses: the bottom cube of a stack, cubes in shadow, white against a white wall.

2 Colour check

HSV of the box centre. An unsaturated "red" or "green" is relabelled white or black; a bright "black" box is rejected.

3 Duplicate + side-face merge

Same-colour boxes side by side are one turned cube: the most confident box is kept. Boxes with width / height < 0.6 are side faces.

4 Stack grouping

Boxes in one column with similar width form a stack, sorted bottom → top by height in the image. Side faces join the column they touch.

5 Hidden-cube probe

One cube-step below each stack: if every cell of a 3×3 grid matches one cube colour, a cube is inferred: dashed box, "(geometric reasoning)".

6 Reading

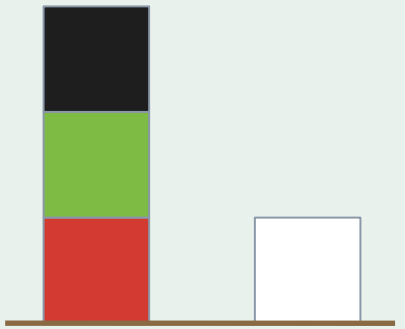
Stacks bottom → top plus single table cubes, e.g. red>green>black | white. Passed to the order check against the goal.

When is the VLM called? Only on an anomaly

Trigger	Meaning	Example (goal r>g>b>w)
Wrong order	A stacked cube is not the goal cube for its position	stack r > g > w
Cube not seen	A goal cube is detected nowhere: YOLO probably missed it	white not found
Gap in a stack	Spacing in a stack far larger than usual: a cube may be hidden	r > [gap] > b
v key	Manual check by the worker	any scene

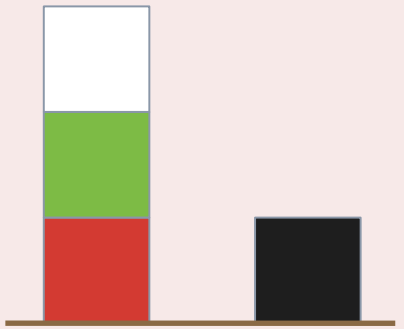
Not an anomaly: a complete stack, or a stack that is correct so far with the remaining cubes on the table. The code planner answers at once and the panel says "VLM not called: no anomaly against the goal".

No anomaly



in progress → code planner:
1. Put WHITE on BLACK

Anomaly



wrong order → VLM checks:
1. WHITE to the table
2. BLACK on GREEN
3. WHITE on BLACK

VLM check: LOOK → GROUND → PLAN

1 LOOK

Read the scene

- Input: raw settled frame, no boxes or YOLO hints (768 px)
- Output (JSON schema): stacks [[bottom … top]], table […]
- Rejected if a colour is listed twice or a confident YOLO cube (≥ 0.6) is missing
- Re-asked once with the reason, else the YOLO reading is used

2 GROUND

Locate missed cubes

- Only for cubes the VLM found but YOLO did not
- Image padded to a 1000×1000 square, so pixel and 0–1000 coordinates agree
- Output: boxes [{colour, box}]
- Box used only if its size, place and colour fit

3 PLAN + verify

Fix the stack

- Text only: accepted reading + goal + worked example
- Output: status, problem, moves [{cube, to}]
- Code replays every move: legal? goal reached? status matches the order check?
- Re-asked up to 2× with the reason, else the code planner

Making a small local VLM reliable

- Ollama REST on localhost:11434, think = false
- Answers forced into a JSON schema; JSON recovered from "thinking" if needed
- Runs in the VLMWorker thread; every settle is queued in order
- A bad answer never blocks: code fallbacks always give steps

Final annotation: where each box comes from

1. **YOLO box**
   Detected box of that colour that fits the stack column best

2. **VLM box**
   GROUND box, if its size, position and colour check out (white: size + position only)

3. **Colour fit**
   Cube-sized blob of that colour where the cube must be

4. **Estimate**
   One cube-step above / below the neighbour, step measured from the stack

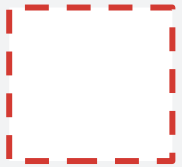
Used in order: the first that works wins.

How boxes are drawn



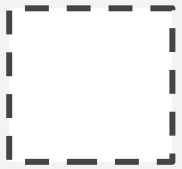
S1-2 green

YOLO detection



S1-1 red (geometric reasoning)

hidden-cube probe



S1-4 white (VLM)

found only by the VLM

YOLO boxes the VLM rejected (e.g. a laptop corner) are not drawn; they are listed in result.json as ignored_detections.

Outputs for every settle

result/

```
goal/goal_<time>/
   goal_raw.jpg, goal_annotated.jpg, goal.json
settles/settle_<time>/
   1_raw.jpg            clean frame
   2_yolo_only.jpg      plain YOLO boxes
   3_yolo_geometric.jpg + geometric reasoning
   4_final_vlm.jpg      final reading + panel
   result.json
stack_log_vlm_<session>.txt
snapshot_<time>.jpg / _stack.txt  (s key)
```

goal

order bottom → top + source

yolo_only

raw YOLO boxes, bottom → top

detector

YOLO + geometry: stacks, verdict, per-cube source

vlm

called? trigger, reading, raw replies, boxes

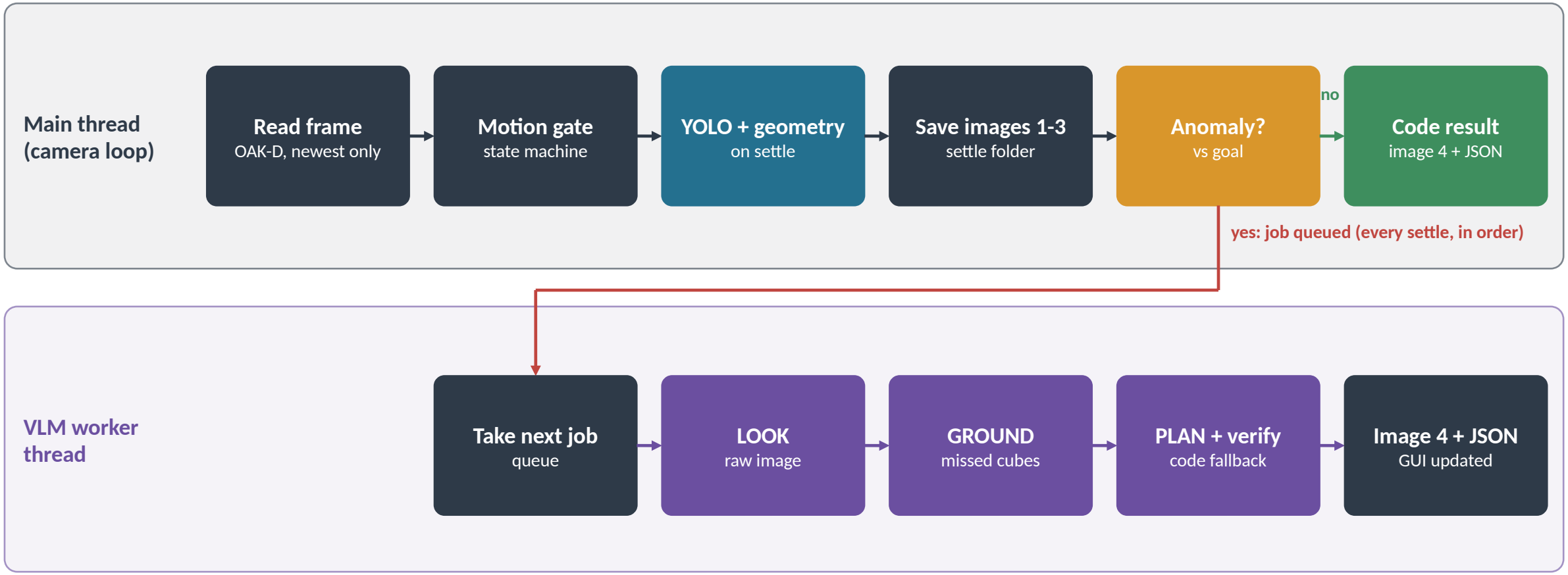
comparison

missed by YOLO / by VLM, placed differently

final

reading, status, instructions, moves, boxes

Runtime: two threads, nothing blocks the camera



Keys

q	quit (pending checks finish first)	**s**	snapshot	**p**	pause	**i**	info line
d	debug detections	**a**	work area	**r**	re-check	**v**	force VLM

Key configuration parameters

Parameter	Default	Effect
`VLM_MODEL`	`qwen3-vl:8b`	Vision model in Ollama (tag case ignored)
`VLM_ON_ANOMALY_ONLY`	`True`	False = call the VLM on every settle
`VLM_ANNOTATE`	`False`	VLM sees the raw image (True: YOLO boxes drawn)
`VLM_GROUND / NO_COLOUR_CHECK`	`True / ('white',)`	Ask for missed-cube boxes; colours not checked by pixels
`VLM_LOOK_RETRIES / VLM_PLAN_RETRIES`	`1 / 2`	Re-asks with the reason before falling back
`VLM_TRUST_CONF`	`0.6`	YOLO cubes the VLM must not drop
`CONF`	`0.25`	YOLO confidence threshold
`SETTLE_SECONDS / CONFIRM_FRAMES`	`0.7 / 3`	Stillness before YOLO; readings that must agree
`MOTION_AREA_FRAC`	`0.01`	Changed share of the work area that counts as motion
`SIDE_FACE_ASPECT / MERGE_X_GAP`	`0.6 / 0.2`	Side-face width ratio; gap for same-cube merge
`PROBE_CELL_MIN_FRAC`	`0.5`	Hidden-cube probe: colour share per grid cell
`MANUAL_WB_K`	`4000`	OAK white balance lock (K)